ABNEESH SINGH

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Systems & AI Engineer | Compilers, Native Developer Tools, On-Device AI
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EDUCATION

Bachelor of Technology in Computer Science and Engineering Expected 2027
Jagran Lakecity University ◇ *Specialization: AI/ML* *Coursework: DSA, OOP, DBMS, OS, Computer Networks*

SKILLS

Programming Languages	C++20, Python, SQL, JavaScript, TypeScript
Generative AI & Agents	LLM Integration, Multi-LLM Orchestration, AI Agent Workflows, RAG, Embeddings, Tool Calling, Prompt Engineering
Systems & Compilers	LLVM, Abstract Syntax Trees (ASTs), Bytecode VMs, Win32 API, IPC, Asynchronous I/O
Backend & Infrastructure	PostgreSQL, MySQL, SQLite, MongoDB, Supabase, Docker, Git, Next.js, Playwright

SELECTED TECHNICAL PROJECTS

EPL – English Programming Language Jan 2024 – Present
Multi-Backend Compiler & Runtime Ecosystem | C++, Python, LLVM, WebAssembly, VM *GitHub | eplang.me*

- Architected an open-source compiler featuring a lexer, recursive-descent parser, AST transformation pipeline, stack-based Bytecode VM, and multi-target LLVM IR cross-compilation pipeline.
- Implemented corpus-based parity testing across the tree-walk interpreter and bytecode VM, detecting divergent program output and exit codes across supported test suites.
- Built a 22-package ecosystem with a 21-module standard library, automated package manager, and official VS Code extension; grew to **4,000+ verified PyPI downloads**.

NeuroShell – Autonomous Native AI Terminal & Runtime Jun 2024 – Present
Hybrid Terminal Host & On-Device AI Engine | C++20, Python 3.12, ONNX Runtime, Win32 *GitHub*

- Architected a native C++20 terminal engine with Win32 ConPTY integration, ANSI VT decoders, and non-blocking process execution managed through Win32 Job Objects, communicating with a Python AI daemon via JSON-RPC 2.0 over Named Pipes and SHM.
- Deployed an INT8-quantized, 270M-parameter EmbeddingGemma 2 text/code model through ONNX Runtime for local semantic retrieval, paired with dynamic TF-IDF and Levenshtein predictive ghost-text command completion.
- Engineered an AST error auto-fixer, an inline Viewport Data Loss Prevention (DLP) secret masker, and a multi-layer command-safety pipeline with an SQLite transaction log (WAL mode, FTS5) for blast-radius analysis and rollback.

IRA – Asynchronous Voice AI Assistant Jul 2024 – Present
Real-Time Voice Pipeline & Local Cognitive Engine | Python, Rust FFI, asyncio, Speech Processing *GitHub*

- Built an asynchronous voice pipeline combining offline wake-word detection, voice activity detection (Silero VAD), Gemini Live speech-to-speech, fallback STT/TTS, and real-time barge-in interruption handling.
- Unified voice capabilities through a shared tool registry normalizing signatures across asynchronous executors, paired with an in-memory entity knowledge graph and native Rust engine (FFI) for context retrieval.

Syting – Real-Time Social Platform & Calling Engine Aug 2025 – Present
PostgreSQL Security, Realtime Messaging & Calling | Flutter, Supabase, PostgreSQL, WebRTC

- Architected a PostgreSQL backend with 80+ migrations, enforcing match lifecycle transitions, server-side recipient derivation, and private profile masking via security-definer database triggers and Row-Level Security (RLS).
- Redesigned Supabase Realtime delivery around recipient-keyed channels instead of per-match subscriptions to support a user-centric model, integrating WebRTC audio/video calling with automated connection recovery.

ASWorks – Digital Creator & Developer Marketplace Aug 2025 – Present
Full-Stack Digital Marketplace & Invoicing Engine | Next.js 15, TypeScript, MongoDB, Stripe *asworks.studio*

- Built a digital software marketplace (asworks.studio) using Next.js 15 Server Components and MongoDB, implementing idempotent webhook processing for Stripe and Razorpay payment events.
- Configured multi-tier testing pipelines spanning unit, integration, and Playwright E2E suites for checkout flows, expiring signed download URLs, and automated GST invoice generation.

CERTIFICATIONS & CREDENTIALS

- **Anthropic**: Claude Code in Action (2026) ◇ **KodeKloud**: AWS EC2 Fundamentals, Cloud Computing Fundamentals